

PROJECT DOCUMENTATION

Hospital Operations and Patient Analytics Dashboard

Module 8 Project Documentation Report

Item	Details
Project	Hospital Operations and Patient Analytics Dashboard – Group 2
Prepared By	Sri Venkateswara Reddy Yakkanti
Technology	Tableau Desktop / Tableau Public, Python, Pandas, NumPy
Tableau Workbook	MedTrack_DV.twb
GitHub Branch	Yakkanti-Sri-Venkateswara-Reddy
Documentation Scope	Dashboard, KPI, data-processing and delivery documentation

Academic Project Submission • Hospital Operations & Patient Analytics

Document Control

Item	Details
Project Title	Hospital Operations and Patient Analytics Dashboard
Project Group	Group 2
Prepared By	Sri Venkateswara Reddy Yakkanti
Module	Module 8 – Documentation and Project Delivery
Dashboard Areas	Hospital Overview, Patient Flow, Department Analytics, Resource Utilization
Tableau Workbook	MedTrack_DV.twb
GitHub Branch	Yakkanti-Sri-Venkateswara-Reddy

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1. Project Overview

Project purpose

The Hospital Operations and Patient Analytics Dashboard is a Tableau-based operational analytics solution that converts hospital admission, occupancy, length-of-stay, readmission, department and resource information into interactive views. The objective is to support monitoring of patient flow, department performance, bed utilisation and hospital capacity.

Project scope

The implemented workbook contains four dashboard areas: Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. The dashboards use interactive filters and navigation to move from high-level hospital KPIs to patient, department and resource-level analysis.

Implementation evidence

The supplied Tableau workbook defines four dashboards named **HospitalOverview**, **PatientFlow**, **Department Analytics** and **Resource Utilization**. The Department Analytics dashboard shown by the project uses the displayed date range 12/20/2024–12/31/2025 with Gender, Severity and Specialty filters.

Key objectives

Objective	Implementation focus
Hospital monitoring	Admissions/case volume, occupancy, average LOS and readmission KPIs
Patient-flow analysis	Admissions, discharges, transfers, case type and readmission views
Department analysis	Case volume, average LOS, occupancy, readmission and efficiency comparisons
Resource utilisation	Bed capacity, bed utilisation and staffing/resource distribution
Interactive analysis	Date, gender, severity, specialty and dashboard navigation controls

2. Dataset and Data Model

Data source used by Tableau

The supplied workbook connects to the analytical Excel source **Hospital_Dataset_Final_With_KPIs.xlsx**, using the table **Hospital_Data**. The Tableau source definition contains 43 fields, including patient/case identifiers, admission and discharge information, clinical/administrative dimensions, revenue and KPI-related measures.

Important fields

Field	Role / purpose
Case_No	Unique case identifier used for distinct admission/case counts
Hospital_ID	Hospital-level identifier
Specialty_Admission / Specialty	Department or specialty analysis
Admission_Date	Time filtering and trend analysis
Admission_Time / Discharge_Time	Admission/discharge operational detail
LOS_Admission / LOS_Generated	Length-of-stay measures
Readmitted / Readmission_Flag	Readmission analysis
Gender / Nationality	Demographic analysis
Severity / CMI Value	Case complexity and severity analysis
Insurance/Payer / InsurancePlanName	Payer analysis
Case type / Surgical Mix / Payer Mix	Case and operational segmentation
Revenue	Financial analysis
Total_Beds / Occupied_Beds / Available_Beds	Capacity and bed utilisation
Occupancy_Rate	Occupancy analysis
Doctors_Count / Nurses_Count / Other_Staff_Count	Staffing analysis

Dataset scale visible in the Tableau source definition: the Excel relation is configured over the A1:AQ5001 range, with a header row.

3. Data Cleaning and Transformation

Preparation approach

The project preparation workflow focuses on converting the integrated hospital data into consistent Tableau-ready measures. The documented project process includes duplicate handling, missing-value handling, standardisation of department/specialty categories, date preparation, and KPI engineering.

KPI engineering logic verified from the project notebook and Tableau workbook

Metric	Verified logic
Total Admissions / Cases	Distinct count of Case_No (COUNTD in Tableau).
Readmission Rate	Count of Readmitted = Yes divided by records with Readmitted = Yes or No.
Bed Utilization Rate	SUM(Occupied_Beds) / SUM(Total_Beds).
Average Occupancy Rate	AVG(Occupancy_Rate).
Hospital Total / Occupied / Available Beds	FIXED by Hospital_ID using MAX of the corresponding bed field.
Department Efficiency Score	Python notebook uses 40% Occupancy Score + 30% LOS Score + 30% Readmission Score after min-max normalisation; higher occupancy is treated as better, while lower LOS and lower readmission are treated as better.

Note: the workbook contains repeated calculated fields for some KPIs across worksheets. The formulas above are taken from the supplied Tableau XML/notebook rather than inferred.

4. KPI Definitions and Values

Dashboard KPI snapshot

TOTAL CASES	AVERAGE LOS	OCCUPANCY RATE	READMISSION RATE
7,172	1.12 Days	64.90%	12.00%

KPI	Definition	Business purpose
Total Cases	Distinct case/admission events represented in the selected dashboard context.	Monitor patient volume and workload.
Average LOS	Average length of stay represented by the dashboard.	Understand patient throughput and bed demand.
Occupancy Rate	Average/selected occupancy measure shown by the dashboard.	Monitor bed capacity pressure.
Readmission Rate	Percentage of known Yes/No readmission records that are Yes.	Monitor repeat-admission patterns.
Bed Utilization Rate	Occupied beds divided by total beds.	Support capacity and bed-allocation analysis.
Department Efficiency Score	Weighted score combining occupancy, LOS and readmission performance.	Compare department operational performance.

Dashboard context note: the 7,172 Total Cases value is the value visibly displayed on the supplied Department Analytics dashboard for the shown filter context (12/20/2024–12/31/2025). The KPI values 64.90%, 1.12 days and 12.00% are also visibly displayed in the project dashboard/KPI artefacts.

5. Dashboard Guide

Hospital Overview

Provides a hospital-level operational summary. The workbook includes KPI sheets for admissions, occupancy, average LOS and readmission, together with admission trend, admissions by department, gender comparison and occupancy trend views.

Patient Flow

Focuses on patient movement and flow-related analysis. The workbook contains admission trend, discharge KPI, transfer KPI, readmission, patient cases by case type and patient transfers by destination views.

Department Analytics

Provides specialty/department-level comparison using case volume, average LOS, occupancy, readmission and efficiency-related views. The supplied dashboard visibly includes Total Cases, Average LOS, Occupancy and Readmission KPI cards, Admissions by Department, Efficiency Ranking by Department, Case Count and department-performance/occupancy tables.

Resource Utilization

Provides capacity and staffing analysis. The workbook includes bed capacity, total/occupied/available bed KPIs, bed utilisation, doctors, nurses, staffing mix, staff by department and hospital resource load views.

6. Dashboard Content Inventory

Dashboard	Visible / worksheet view	Purpose
Hospital Overview	HO_Admission_KPI	Admission/case KPI
Hospital Overview	HO_Occupancy_KPI	Occupancy KPI
Hospital Overview	HO_LOS_KPI	Average LOS KPI
Hospital Overview	HO_Readmission_KPI	Readmission KPI
Hospital Overview	HO_Admissions_Trend	Admission trend
Hospital Overview	HO_AdmissionsByDept	Admissions by specialty/department
Hospital Overview	HO_GenderComparison	Gender comparison
Hospital Overview	HO_OccupancyTrend	Occupancy trend
Patient Flow	PF_AdmissionTrend	Admission trend
Patient Flow	PF_Discharges_KPI	Discharge KPI
Patient Flow	PF_Transfers_KPI	Transfer KPI
Patient Flow	PF_Readmission	Readmission analysis
Patient Flow	Patient Cases by Case Type	Cases by case type
Patient Flow	Patient Transfers by Destination	Transfers by destination
Department Analytics	DA_TotalCases	Total cases KPI
Department Analytics	DA_DepartmentPerformance	Department performance table
Department Analytics	Efficiency Ranking by Department	Efficiency/LOS comparison
Department Analytics	Case Volume by Department	Case count by specialty
Department Analytics	Occupancy Rate by Department	Occupancy by specialty
Resource Utilization	RU_BedCapacity	Bed capacity by specialty
Resource Utilization	RU_BedUtilization	Bed utilisation rate
Resource Utilization	RU_Doctors / RU_Nurses	Staff count KPIs
Resource Utilization	RU_StaffingMix / StaffByDept	Staff distribution
Resource Utilization	Hospital Resource Load / Bed_Utilization	Resource and utilisation analysis

7. Analytics Methodology

End-to-end workflow

Data Collection → Data Preparation → KPI Engineering → Dashboard Development → Dashboard Integration → Testing & Validation → Documentation & Delivery

Data Preparation

The source data is prepared for analytical use through numeric conversion of bed/resource measures, handling of readmission categories, calculation of occupancy-related measures and grouping/aggregation by specialty, hospital and case. The project notebook also creates department-level metrics and normalised performance scores.

KPI Engineering

The project uses distinct case counts, average LOS, occupancy, readmission and bed-utilisation measures. Department efficiency is generated from three normalised components: occupancy, LOS and readmission.

Dashboard Development

Tableau worksheets are assembled into four dashboard areas. The workbook includes navigation and filters so users can inspect the same operational data from different perspectives.

Operational analytics value

The resulting dashboards support hospital workload monitoring, patient-flow review, department comparison, bed-capacity monitoring and staff/resource analysis. The solution is intended for operational analytics rather than clinical diagnosis or treatment.

8. Dashboard Integration

Filters and interaction controls

The supplied Department Analytics dashboard visibly provides an Admission Date range, Gender filter, Severity filter and Specialty filter. These controls allow the user to change the analytical context without rebuilding the views.

Navigation

The dashboard suite uses a common navigation structure linking Hospital Overview, Patient Flow, Department Analytics and Resource Utilization.

Cross-dashboard analysis

The integrated design lets users move from hospital-level KPIs to patient-flow details, then to department performance and finally to resource/capacity views. This reduces the need to interpret admissions, occupancy, LOS and staffing measures separately.

Department Analytics example

Displayed element	Observed value / content
Date context	12/20/2024 – 12/31/2025
Gender	(All)
Severity	(All)
Specialty	(All)
Total Cases	7,172
Average LOS	1.12 Days
Occupancy	64.90%
Readmission	12.00%

9. Testing and Validation

Test area	Validation approach	Expected result
KPI validation	Recalculate selected KPI logic and compare with dashboard values.	Dashboard KPI values match the implemented logic for the selected context.
Filter testing	Apply date, gender, severity and specialty filters.	Relevant views update to the selected context.
Department analysis	Cross-check case volume, LOS, occupancy and readmission views.	Department comparisons remain consistent.
Resource analysis	Cross-check total/occupied/available beds and utilisation views.	Capacity measures remain internally consistent.
Navigation testing	Use dashboard navigation controls across the four areas.	Users can move between dashboard areas without broken navigation.
Workbook packaging	Open the Tableau workbook and verify dashboards/data dependencies.	Workbook opens with required analytical content.

10. Project Structure and Technical Stack

Area	Project artefacts / technology
Data	Hospital_Dataset_Final_With_KPIs.xlsx / Hospital_Data
Python processing	generate_hospital_kpis_py.ipynb, Pandas, NumPy
Visualization	Tableau Desktop / Tableau Public
Workbook	MedTrack_DV.twb
Version control	GitHub branch: Yakkanti-Sri-Venkateswara-Reddy
Documentation	PDF project documentation

11. Final Deliverables and Limitations

Final deliverables include the Tableau workbook, KPI/data-processing notebook and project documentation. The dashboard is an operational analytics solution; its results depend on the quality and consistency of the underlying hospital dataset. KPI values should always be interpreted in the context of the filters selected in Tableau.

Source and verification note: this report was prepared from the supplied Tableau workbook, KPI-generation notebook, the supplied dashboard/documentation artefact and the dashboard values visible in the conversation. Where an implementation detail was not verifiable, it has not been invented.